Republic of the Philippines
University of Southeastern Philippines
Iñigo St., Obrero, Davao City 8000
COLLEGE OF ENGINEERING

PROGRAM STRUCTURE

DEGREE PROGRAM: **Bachelor of Science in Electronics Engineering**

Primary Job Roles students are prepared for in the program	**Telecommunication Engineer, Biomedical Engineer, Computer Network Engineer, Practicing Electronics Engineer/Building Management System Engineer/Contractor**
Secondary Job Roles students are prepared for in the program	**Academician, Technical Support Engineer, Program Analyst, Instrumentation & Control Engineer, Satellite Communication Engineer, Broadcast Engineer, Mechatronics Engineer, Manufacturing Engineer, Engineering Technician, Electronics Technician**

Primary and Secondary Job Roles provide a mental anchor to the identification of program outcomes you will be having for your program. This will allow you to check for completeness in your program outcomes.

Current CMO Guiding Program: CMO 101, Series of 2017

In accordance with the pertinent provisions of Republic Act (RA) No. 7722, otherwise known as the "Higher Education Act of 1994," in pursuance of policies, standards and guidelines in the establishment of an outcomes-based education (OBE) system in higher education institutions offering engineering programs as advocated under CMO 37 s. 2012 and policy-standard to enhance quality assurance (QA) in Philippine higher education through an outcomes-based and typology-based QA as advocated under CMO 46 s. 2012, and by virtue of Commission en banc Resolution No. 788-2017 dated October 24, 2017 the following policies and standards are hereby adopted and promulgated by the Commission.

This CMO integrates the establishment of an outcomes-based Based on the Guidelines for the Implementation of CMO No. 46 series of 2012 and CMO 37 s. 2012, this PSG implements the shift to outcomes-based education (OBE) leading to competency-based standards. It specifies the "core competencies" expected of Bachelor of Science in Electronics Engineering (BSECE) graduates "regardless of the type of Higher Education Institutions (HEI) they graduate from." However, in recognition of outcomes-based education and the typology of HEIs, this PSG also provide ample space for HEIs to innovate in the curriculum in line with the assessment of how best to achieve learning outcomes in their particular contexts and their respective missions.

Program Description (CMO 101 Series of 2017):

Definition
ELECTRONICS is the science dealing with the development & application of devices and systems involving the flow of electrons, or other carriers of electric charge, in vacuum, in gaseous media, in plasma, in semiconductors, in solid-state and/or in similar devices, including but not limited to, applications involving optical, electromagnetic & other energy forms when transduced or converted into electronic signals.

Degree Name
In consonance with the field of specializations of the Electronics Engineering Profession stipulated in section 5, article 1 of RA 9292 also known as Electronics Engineering Law, the degree program herein shall be called BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING (BSECE).

Nature & Scope of the practice of an ECE graduate
The scope and nature of practice of the Electronics Engineer shall embrace and consist of any work or activity relating to the application of engineering sciences and/or principles to the investigation, analysis, synthesis, planning, design, specification, research and development, provision, procurement, marketing and sales, manufacture and production, construction and installation, tests/measurements/control, operation, repair, servicing, technical support and maintenance of electronic components, devices, products, apparatus, instruments, equipment, systems, networks, operations and processes in the fields of electronics, including communications and/or telecommunications, information and communications technology (ICT), computers and their networking and hardware/firmware/software development and applications, broadcast/broadcasting, cable and wireless television, consumer and industrial electronics, electro optics/photonics/opto-electronics, electro-magnetic, avionics, aerospace, navigational and military applications, medical electronics, robotics, cybernetics, biometrics and all other related and convergent fields; it also includes the administration, management, supervision and regulatory aspects of such works and activities; similarly included are those teaching and training activities which develop the ability to use electronic engineering fundamentals and related advanced knowledge in electronics engineering, including lecturing and teaching of technical and professional subjects given in the electronics engineering (RA 9292, 2004).

VISION
A PREMIER UNIVERSITY IN THE ASEAN REGION

MISSION
USeP shall produce world-class graduates and relevant research and extension through quality education and sustainable resource management.

CORE VALUES
USeP is a community of scholars that values:
U – Unity
S – Stewardship
E – Excellence
P – Professionalism

Program Educational Objectives		Mission-Vision and Core Value Alignment						Reference Source or CMO Provision Compliance
		Vision	Mission	Core Value				
				U	S	E	P	
PEO1	Demonstrate expertise in the field of Electronics Engineering	√	√		√	√	√	USEP Brochure, CMO 101
PEO2	Actively participate in professional communities to nourish ever-developing careers	√	√	√	√		√	USEP Brochure, CMO 101
PEO3	Demonstrate gender sensitive leadership and commitment to work in diverse cultural and interdisciplinary backgrounds	√	√	√		√	√	USEP Brochure, CMO 101
PEO4	Exhibit globally competitive skills	√	√		√	√	√	USEP Brochure, CMO 101

PERFORMANCE INDICATORS

Complete the performance indicators for each of the outcomes that we are able to generate during the workshop. The starting statements are not yet complete; they are just starting phrases to guide you in your performance indicators formulation

Institutional Graduate Attributes:

IGA1 – Critical Analysis
IGA2 – Leadership and Teamwork
IGA3 – Service-Oriented
IGA4 – Professional Competence
IGA5 – Communication
IGA6 – Life-Long Learning

Institutional Graduate Attribute	Program Outcomes		Performance Indicators	
IGA1	**a**	Apply knowledge of mathematics and sciences to solve engineering problems;	1	Apply concepts in mathematics to solve engineering problems
IGA4			2	Apply principles in chemical sciences to solve engineering problems.
IGA6			3	Apply principles in physical sciences to solve engineering problems
IGA1, IGA4	**b**	Design and conduct experiments, as well as to analyze and interpret data	1	Perform laboratory experiments as prescribed by standard procedures.
IGA5, IGA6			2	Design an experiment to achieve a specific objective
IGA1, IGA2			1	Develop an engineering solution that have various constraints as factors

IGA3, IGA4 IGA5, IGA6	**c**	Design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards.	2	Design a system or process based on various constraints
IGA2	**d**	Function on multidisciplinary teams	1	Shows tact in dealing with people he/she comes in contact with.
IGA3			2	Capable of learning from and listening to co-members
IGA6			3	Shows respect and courtesy in dealing with peers and superiors.
IGA1	**e**	Identify, formulate, and solve engineering problems	1	Interpret statement problems into mathematical expressions.
IGA4			2	Apply concepts, theories, and principles in solving engineering problem(s)
IGA6			3	Solve specific engineering problem related to discipline.
IGA4	**f**	Understand professional, social and ethical responsibility	1	Internalizes the professional code of ethics for specific discipline.
IGA5			2	Applies the principles of professional ethical standard and social responsibility to actual practice
IGA6				
IGA4	**g**	Communicate effectively	1	Articulate the convention and style in oral communication that conveys complete thought and information.
IGA5			2	Articulate the convention and style in written communication that conveys complete thought and information.
IGA6			3	Demonstrate proficiency in using graphs, tables, drawings, symbols, etc. applicable in the discipline.
IGA1, IGA3	**h**	Understand the impact of engineering solutions in a global, economic, environmental, & societal context through broad education	1	Relate engineering solutions to global, economic, environmental, and societal context.
IGA4, IGA6			2	Describe engineering solutions in application to global, economic, environmental, and societal context.
IGA1, IGA2	**i**	Recognize the need for, and an ability to engage in life-long learning	1	Demonstrate ability to learn independently.
IGA3, IGA4			2	Shows appreciation and understanding of the engineering profession.
IGA5, IGA6			3	Understand the linkage or connection between his/her task to previous, intervening and subsequent tasks.
IGA2, IGA3	**j**	Apply knowledge of contemporary issues	1	Identify the socio-economic, political, and environmental issues related to the practice of engineering
IGA4, IGA5			2	Explain the possible impact of technology to local, national, and international issues
IGA6			3	Shows leadership skills in managing a multidisciplinary group.
IGA1	**k**	Use techniques, skills, and modern engineering tools necessary for engineering practice	1	Handles the details of the work assigned to him/her.
IGA4			2	Demonstrate the ability to operate the equipment needed in the engineering practice.
IGA6			3	Use modern technology in performing engineering practice.
IGA1, IGA2	**l**	Apply knowledge of engineering and management principles as a member and leader in a team, and to manage projects and in multidisciplinary environments	1	Capable of learning from and listening to co-members.
IGA3, IGA4			2	Shows leadership skills in managing a multidisciplinary group.
IGA5, IGA6			3	Participates in discussions and comes up with sound suggestions to problems.

There are about six (6) fundamental attributes for any professional that belongs to high-capabilities being developed in Higher Education. To check if your program outcomes are complete, you must have a representative outcome to each of the attributes indicated above. The attributes above are minimum; you can add additional attributes and outcomes. It is possible that in one attribute, there are more than one outcome

CURRICULUM MAP

LEGEND

Code	Descriptor	Definition
I	Introductory Course to an Outcome	A formative course to an outcome
E	Enabling Course to an Outcome	A course strengthening further an outcome
D	Demonstrative Course to an Outcome	A course demonstrating an outcome

				Program Outcomes											
Year Level	**Semester**	**Units**	**Courses in the Curriculum**	**a**	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**	**J**	**k**	**l**
FIRST YEAR	**FIRST**	4	Engineering Calculus 1	I				I							
		4	Chemistry for Engineers	I	I			I							
		1	Engineering Drawing and Plans				I			I					
		2	Computer Programming (Object Oriented Programming)		I							I		I	
		1	Integration Course for ECE 1		I							I		I	
		3	Purposive Communication												
		3	Mathematics in the Modern World												
		3	Science, Technology, and Society												
		2	Movement Enhancement												
		3	National Service Training Program 1												
	SECOND	4	Engineering Calculus 2	E				E							
		4	Physics for Engineers	E	I		I	E							
		4	College Physics 2	E	I	I		E							
		1	Computer Aided Drafting						I	I		I		I	
		1	Integration Course for ECE 2		I							I		I	
		3	Understanding the Self												
		3	Ethics												
		2	Fitness Exercises												
		3	National Service Training Program 2												
SECOND YEAR	**FIRST**	3	Differential Equations	E				E							
		3	Engineering Data Analysis	I				I			I				
		3	Electrical Circuits 1	E				E							
		1	Electrical Circuits 1 Lab		E									E	
		3	Electronic Devices and Circuits	E				E							
		1	Electronic Devices and Circuits Lab		E	I									
		1	Fundamentals of Scientific Computing		I							E		E	
		1	Integration Course for ECE 3		E							E		E	
		3	The Life & Works of Jose Rizal												
		3	Readings in Philippine History												
		2	Physical Activities Towards Health & Fitness I												

				Program Outcomes											
Year Level	Semester	Units	Courses in the Curriculum	a	b	c	d	e	f	g	h	i	J	k	l
		3	Electrical Circuits 2	E				E							
		1	Electrical Circuits 2 Lab		E									E	
		3	Electronic Circuit Analysis & Design	E				E							
		1	Electronic Circuit Analysis & Design Lab		E	E									
		3	Principles of Communication Systems	E				E							
		1	Principles of Communication Systems Lab		E	E								E	
		3	Advanced Eng'g Mathematics for ECE	E				E							
		1	Advanced Eng'g Mathematics for ECE Lab		E									E	
		1	Integration Course for ECE 4		E		E					E			
		3	Engineering Economics	E					I		I		I		I
		3	Art Appreciation												
		2	Physical Activities Towards Health & Fitness II												
THIRD YEAR	FIRST	3	Electronic Systems and Designs	E				E			E				
		1	Electronic Systems and Designs Lab		E	E									
		3	Signals, Spectra, Signal Processing	E				E							
		1	Signals, Spectra, Signal Processing Lab		E									E	
		3	Modulation & Coding Techniques	E				E							
		1	Modulation & Coding Techniques Lab		E									E	
		3	Logic Circuits Design & Switching Theory	E		E		E							
		1	Logic Circuits Design & Switching Theory Lab		E					E					
		4	Electromagnetics for ECE	E				E							
		3	Contemporary World									E			
		3	General Education Elective 1												
	SECOND	3	Data Communications	E				E							
		1	Data Communications Lab		E									E	
		3	Transmission Media, Antenna System & Design	E				E		E					
		1	Transmission Media, Antenna System & Design Lab		E	E								E	
		3	Feedback and Control Systems	E				E							
		1	Feedback and Control Systems Lab		E									E	
		3	Microprocessor & Microcontroller Systems	E				E							
		1	Microprocessor & Microcontroller Systems Lab		E	E				E				E	
		3	Methods of Research						E	E	E		E		E
		3	Elective 1	E								E			
		1	Elective 1L		E							E			
		3	General Education Elective 2												

				Program Outcomes											
Year Level	Semester	Units	Courses in the Curriculum	a	b	c	d	e	f	g	h	i	J	k	l
FOURTH YEAR	FIRST	3	Material Science and Engineering	E				E							
		3	Environmental Science and Engineering						E		E		E		E
		3	Capstone Project	D	D	D		D	D	D	D		D	D	D
		3	ECE Laws, Contracts, Ethics Standards & Safety						E			E	E		
		1	Industrial Plant Visits and Seminars				E		E	E		E		E	
		1	Free Elective							E			E		
		3	Elective 2	E								E			
		1	Elective 2L		E							E			
		3	Engineering Management				E		E	E	E		E		E
		3	Technopreneurship				E		E	E					E
		3	General Education Elective 3												
FOURTH YEAR	SECOND	3	Thesis/Capstone Project	D	D	D		D	D	D	D	D	D	D	D
		2	Correlation Course for ECE	D				D				D			
		2	On-the-Job Training (480 HOURS)				D		D	D		D	D	D	

REFLECTION FORM

After going through this exercise of planning for your course, give yourself a 15-minute break and reflect on the following questions below. For each of the question, indicate the rating

5 Very Likely
4 Less Likely
3 Cannot Tell
2 Not Likely
1 Definitely Not

	5	4	3	2	1
Do you think that your course now has a better purpose to the professional or personal life of the student?	√				
Do you think that your course now has a better alignment to the intentions of the degree program or the intentions of the institution?	√				
Do you think that your course planning technique has gone better this time than before?	√				
Do you think you have designed a course now with a better plan as to how students will progress inside the course?	√				
Do you think your topics will be better anchored to a specific purpose/s than before?	√				
Do you think you are planning your courses now for students to succeed and achieve capabilities?	√				
Do you think that if this course will be properly implemented, your students will be developed better?	√				
Do you think that if you are given training on better instructional delivery, this course is exciting to teach?	√				

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